Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 13)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 1 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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0 1 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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0 2 . 1 Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

A Children inhale more lead vapour ☐

B Children pick up dust / flakes and suck their fingers ☐

C Lead is more soluble in children's blood ☐

D Children spend more time near roads ☐

0 2 . 2 Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

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0 2 . 3 Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

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0 2 . 4 Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

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0 2 . 5 Explain why heavy metals are described as neurotoxins.

[1 mark]

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0 3 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 3 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 3 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 4 . 1 Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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0 4 . 2 Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

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0 4 . 3 Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

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0 4 . 4 Describe **two** environmental impacts of solar PV.

[2 marks]

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0 5 Table 1 shows the mean annual soil loss measured on four fields of one farm. Soil forms on this farm at about 1 t ha^{-1} yr^{-1}.

Table 1

Field	Slope	Land use	Mean annual soil loss / t ha^{-1} yr^{-1}
A	2%	Permanent grassland	0.8
B	9%	Maize, ploughed up and down the slope	24.0
C	9%	Maize, contour ploughed	9.0
D	1%	Spring barley, soil left bare over winter	6.5

0 5 . 1 Use Table 1 to identify the fields that are suffering **accelerated** soil erosion, and explain your answer.

[2 marks]

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0 5 . 2 Calculate the percentage reduction in soil loss achieved by changing field B to contour ploughing.

[2 marks]

Answer ..... %

0 5 . 3 Explain how each of the following reduces erosion by water.

[3 marks]

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Contour ploughing .....

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Terracing .....

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Rows of stones .....

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0 5 . 4 Field D is exposed to strong winds. Explain **four** methods, other than planting hedgerows, that would reduce wind erosion there.

[4 marks]

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0 5 . 5 Describe **four** impacts of the eroded soil **away from** the fields where it is lost.

[4 marks]

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0 6 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 6 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 7 Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

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0 7 . 1 Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

0 7 . 2 Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

0 7 . 3 State **two** changes that would decrease the cut-off ore grade.

[2 marks]

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0 8 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 8 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

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0 8 . 2 Explain why readings should be taken with the same wind direction.

[1 mark]

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0 8 . 3 Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

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0 8 . 4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

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0 9 . 1 Explain **three** features of fossil fuels that made them ideal for the development of industrial society.

[3 marks]

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0 9 . 2 75 t of aviation fuel carries a Boeing 747 and its 400 passengers from London to New York. Calculate the mass of fuel used per passenger, in kg.

[2 marks]

Answer _____ kg

0 9 . 3 Explain why the total quantity of fossil fuels in existence is much greater than the quantity counted as **reserves**.

[3 marks]

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0 9 . 4 Cheap North Sea gas contributed to the closure of the UK's deep coal mines, although large coal deposits remained. Explain why those mines cannot now be reopened.

[2 marks]

0 9 . 5 Explain **four** environmental impacts of the **extraction** of fossil fuels, as distinct from burning them.

[5 marks]

15

1 0 . 1 Ozone depletion is often described as an environmental problem that has been tackled successfully. Compare the effectiveness of the methods used to restore the ozone layer with the effectiveness of attempts to tackle global climate change, and explain the reasons for the difference.

[9 marks]

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Extra space

1 0 . 2 Name the international agreement of 1985 that preceded the Montreal Protocol and state its main weakness.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS